up. So, there was a natural move towards that direction. And I think that the RISC-V movement, thanks to such a deal, is going to accelerate. The advantage that the Arm ecosystem has and will continue to have for the next several years is really a huge ecosystem of support that has been established around it. No matter what, it will still take time for any organisation to develop that kind of infrastructure support. It will take a while for anybody to replicate it. If Arm is true to its word of being an open-system architecture, then that is no big deal. But it's always good to be careful since we as companies are at the receiving end of things.

Where all are your R&D teams spread? Is there a team in India too?

Earlier there wasn't but now there is an R&D team in India. We are establishing a footprint in Bengaluru primarily for RF and timing devices that have very good capability. And we have leveraged our partnership with a company called Steradian. We have always used India-based design outfits and worked with companies like Tata Elxsi, HCL, Sankalp, and TCS. We have decided to establish a good footprint in India.

Is there a roadmap for India that you would like to share now, or is it under progress?

For RF, we certainly have a very compelling roadmap. A lot of our millimetre wave activities are currently done in India. By millimetre wave I don't mean sub-6GHz, but millimetre wave that is 28GHz and above. We are also working on some of the timing activities in that area and looking at setting up a stronger presence in Bengaluru for that activity. Radar activity also goes on there.

Are e-bikes and surveillance the two key markets or are there other markets also wherein you see some growth?

There are other markets too where we see a growth in. Bio-sensing is one. Due to Covid-19, getting low-cost healthcare to people and air-quality monitoring is going to become much more important. Our team is going after a range of customers in IoT, mobility, medical, consumer, infrastructure, and industrial sectors. It is quite exciting.

My view is that India is an untapped market in terms of electronics reach. And the potential is just enormous. So, in addition to the R&D, for which we have established a footprint, we have plans to work with emerging companies like Ather, besides the established where we have already established footprint. The reach is going to be quite significant and, if India sets its mind to build the white boxes themselves, I think it will be a massive gamechanger for us. If it starts to do some of the things that are done in Taiwan, including the product design specs, then it will propel us to the next level.

Do you see IoT as a buzzword or a real market opener?

That's a good question. To me, any edge or endpoint device is fundamentally IoT. Are more devices and more sensors getting connected across the globe? Yes, absolutely. Without a doubt, we are marching towards one trillion connected devices in a not too distant future. It doesn't matter to me whether you call it IoT or connected sensors. The trend is already there.

For us to capitalise on that market, it boils down to having all the intellectual property that is needed for it. Companies will not be able to go very broad in this area, and that is the fundamental challenge in IoT. There's no single company that is going to dominate every vertical, because not only we need to have the sensor solution, we also have to be able to interface it to something else in order to get it to operate. If it's a smart home, whether it's a Google Nest thermostat or something else, it needs to interface with another device and the software. So, you have to decide the verticals you are going to go after. Like in India, we are going after smart metering. Motor control is another great vertical that we are pursuing.

Any particular geographies like Europe or the USA that have taken lead in industrial IoT?

For industrial IoT, the areas that are doing well are Japan, Europe, and then the USA. China has an interesting growth story for a variety of reasons, including the fact that they came out of Covid-19 crisis much earlier than a lot of other regions in the world. One of the areas that they do continue to surprise a lot of people is of AI. The dialects in China (certainly not as complex as in India) are vastly different. They have AI companies like iFLYTEK that are really advanced when it comes to voice recognition-based AI technologies. So, China is probably far ahead of India when it comes to industrial IoT, although India is starting to pick up.

Most of the advanced robotics is largely happening in regions such as China, Japan, and the USA. The traditional base of industrial customers has largely been European, Japanese and the American to some extent. However, China has been increasingly very aggressive in terms of industrial companies that adopt new products quickly. The advantage that they have is that they move extremely fast and don't require the 10/15/20-year lifetime reliability cycles, which the traditional players require. They understand that most of these products would not last more than 7-8 years before you will have to upgrade them to the next version. So, watch that space as there is going to be a big set of disruptions including digitisation, which will get hold of that market over the next 5-10 years and major new players will emerge.

Having seen the entire silicon ecosystem from pretty close, do you think India should go ahead and invest in a semiconductor fab or remain fabless?

India has tried investing in fab earlier as well and this isn't the first time. If we take a look at the history of countries that have been successful in introducing fabs, such as Taiwan who started investing in the late 70s and early 80s, it took thirty years of learning to become what they are now. There is no short-circuiting that process. They have the companies and supplier base, despite being located on a small island, which is a fraction of a vast country like India.

China has repeatedly tried to establish fabs, with far more resources than India, to be successful in the wafer fabrication area. And they are still several decades away from getting there. So, the question is: Do we have the long-term vision to make US$70-80 billion worth of investment? It is critical to identify the vertical after which you want to go, such as communications and enterprise infrastructure, and then start establishing a company that will build the systems.

If it's for 5G, then there is a perfect opportunity with Open RAN systems to establish a company in India that could set up their own telecom infrastructure. Open RAN is hugely disruptive versus traditional telcos. If you do that, then you will naturally start establishing fabless players and others that will support that ecosystem and set up the entire manufacturing system for it. But, without identifying the vertical to pursue, you won't establish a strong footprint. The semiconductor industry in India will be challenged. Huawei became what it is today because they built the boxes, and that set up Hi Silicon, which led to set up other companies. EFY